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Directional Forecasting of Pension Investment Funds

Evidence on the Majority Class Illusion from ARIMA, CNN, and LSTM Models

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Data: TEFAS · Period: April 2021 – April 2026 · Funds: ALZ • AZS • AMZ · 3 seeds × 81 configs / fund

Outline

Literature Review

Classical ARIMA

Box-Jenkins ARIMA remains the dominant benchmark in pension-fund return forecasting. Karadağ Erdemir & Kırkağaç (2024) applied ARIMA to Turkish BES pension funds. Eşsiz & Ordu (2024) applied ARIMA to S&P 500 fund baskets. Strong parametric baseline, blind to non-linear structure.

Evaluation Methodology Gap

Finance DL studies frequently report raw accuracy without specificity, sensitivity or baseline comparison. Majority-Class illusion (Spec=0 or Sens=0) is rarely flagged, leading to overstated predictive performance on imbalanced data. This study addresses this gap explicitly.

Deep Learning in Finance

Fischer & Krauss (2018) showed LSTM networks outperform memory-free models on S&P 500 constituents. Siami-Namini et al. (2019) reported LSTM/BiLSTM improvements over ARIMA on economic time-series. 1D-CNN offers a competitive and computationally cheaper alternative (Van der Burgt, 2023).

Research Contribution

First systematic application of ARIMA + LSTM + 1D-CNN to TEFAS pension funds (ALZ/AZS/AMZ) under a multi-seed, multi-window framework — with Majority-Class detection, Balanced Accuracy and Naïve-Baseline comparison built in from the start.

Motivation — Why BES, Why Directional?

Why Pension Funds?

Turkey's TEFAS platform manages billions in pension assets. ALZ, AZS and AMZ represent three distinct risk profiles (low ±2%, medium ±18%, high ±38%). Directional prediction directly impacts retirement income and portfolio-rebalancing decisions.

Research Gap

Prior work (Karadağ Erdemir & Kırkağaç, 2024; Eşsiz & Ordu, 2024) relies on ARIMA for pension-fund returns. No prior study systematically applies LSTM and 1D-CNN with multi-seed validation to TEFAS pension funds.

Adapted Methodology

Van der Burgt (2023) NASDAQ-100 framework extended to weekly TEFAS data: multi-seed (23/27/98), rolling windows (In∈{2,4,6}, Out∈{1,3,5}), 3 feature sets (full · technical · closing), Majority-Class detection and Naïve-Baseline comparison.

Why Balanced Accuracy?

Data is heavily imbalanced (AZS: 84.85% Up). Raw accuracy is misleading — always predicting Up gives 84.85% but captures no Down signal. Balanced Accuracy = (Sensitivity + Specificity) / 2 penalises the Majority-Class illusion and is the primary metric throughout.

Data & Pension Funds — ALZ / AZS / AMZ

ALZ

Low-Risk

Conservative • Swing ± 2 % • MC: 27 / 27 MC

Allianz Yaşam — Money-market dominant. Almost monotone series.

AZS

Medium-Risk

Mixed • Swing ± 18 % • MC: 9–13 / 27 MC

Mixed equity/debt fund. Moderate volatility, partial signal.

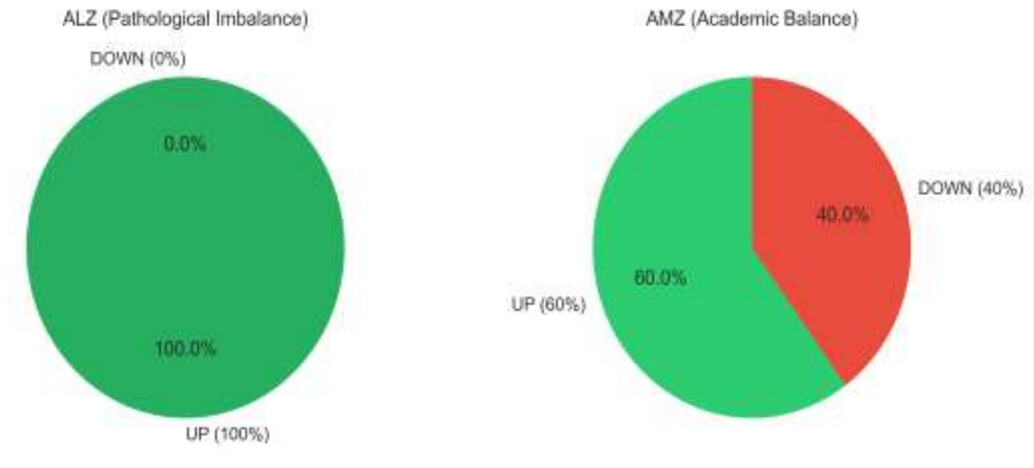
AMZ

High-Risk

Equity-Heavy • Swing ± 38 % • MC: 9–10 / 27 MC

High-volatility equity fund — highest learnable directional signal.

Figure1 : Two-Fund Profile



DATA SUMMARY

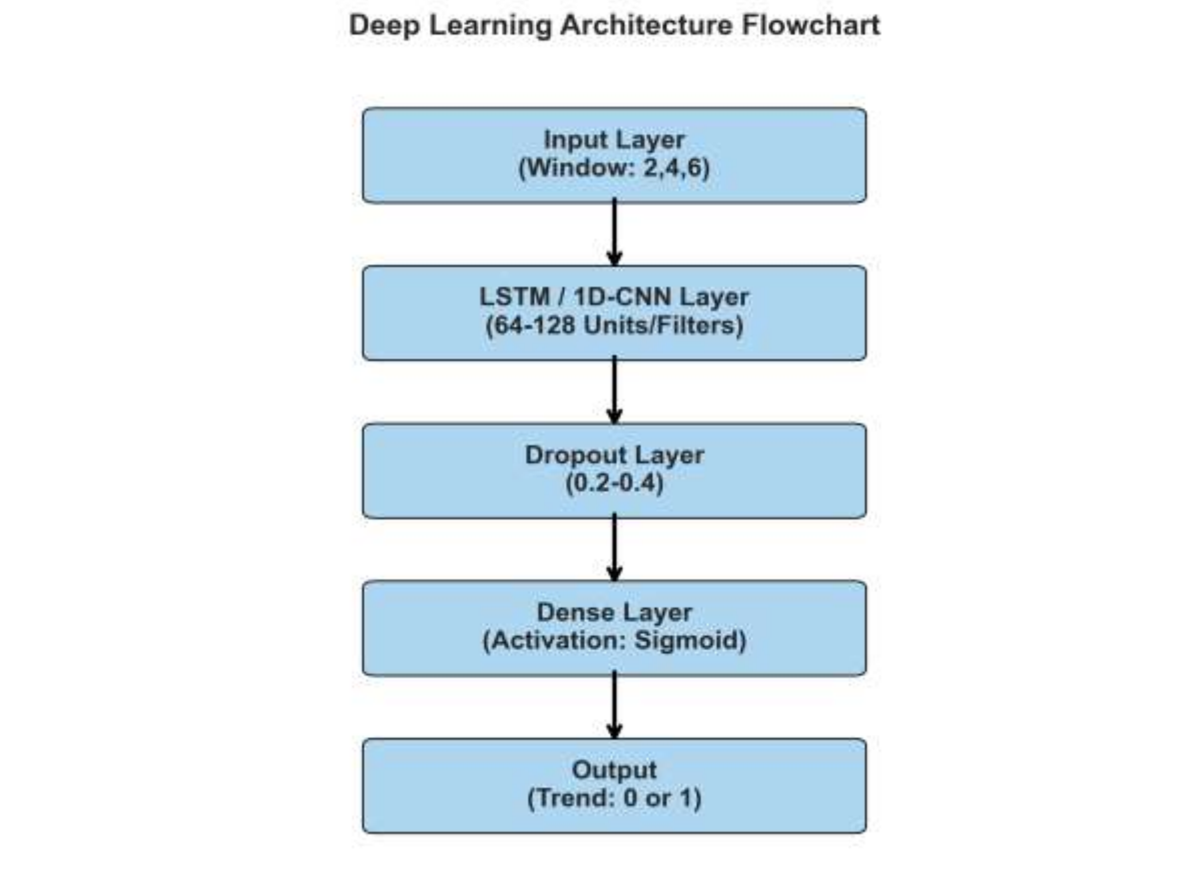
Source: TEFAS (Turkish Electronic Fund Trading Platform)

Period: April 2021 – April 2026 (~260 weekly observations)

Split: 70 % train / 15 % val / 15 % test (chronological)

Target: binary direction (close[t+h] > close[t] ?)

Methodology — Three-Model Pipeline



ARIMA
Box-Jenkins
Grid d∈{0,1,2,3,4,5}, CSS fit
Classical baseline

LSTM
Recurrent
Adam / SGD • ReLU/Tanh
Dropout {0, 0.2, 0.4} •
class_weight

1D-CNN
Convolutional
Filters {32-64-128, 64-128-256}
Kernel 3 • Dense 64 •
class_weight

All models share the same 9 (Input × Output) windows and 3-seed protocol

Search Space — 81 Configurations per Fund

3
Feature Sets
closing • technical • full

9
(In, Out) Windows
(2.4.6) × (1.3.5) weeks

3
Random Seeds
23 • 27 • 98

81
Configs / Fund
3 × 9 × 3

Figure2 : Hardware & Hyperparameter

Specification	Value
Environment	R Keras + TensorFlow 2.21.0
Architecture	LSTM (128) / 1D-CNN (64 Filt)
Optimizer Pool	Adam, SGD, RMSprop
Dropout	0.2, 0.4
Hardware	Core i7-13700H / 32GB RAM
Stop Criterion	Early Stopping (Patience: 5)

FEATURE SETS

closing *— 1 var*
Weekly close only

technical *— 3 vars*
RSI(14) • MACD • Momentum(10)

full *— 7 vars*
Close, RSI, MACD, EMA12, EMA26, Momentum, Volatility

Training: epochs = 50 • batch = 16 • patience = 5 (early stopping) • class_weight balancing • CSS fit for ARIMA

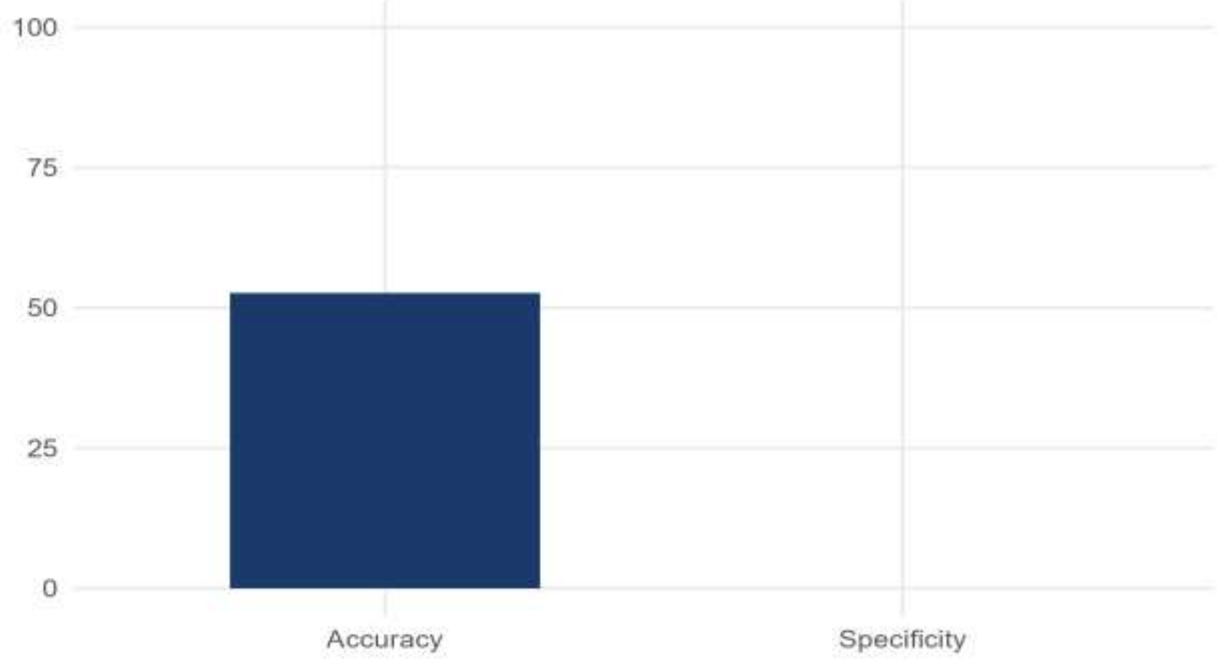
The Majority-Class Illusion

Why a 90 % accuracy can mean a 0 % learner.

27/27 of ALZ low-risk configurations are majority-class — perfect-looking accuracy without any learning.

The Accuracy Illusion in Action

Figure3 : LSTM Closing Accuary



MC DEFINITION (broad)

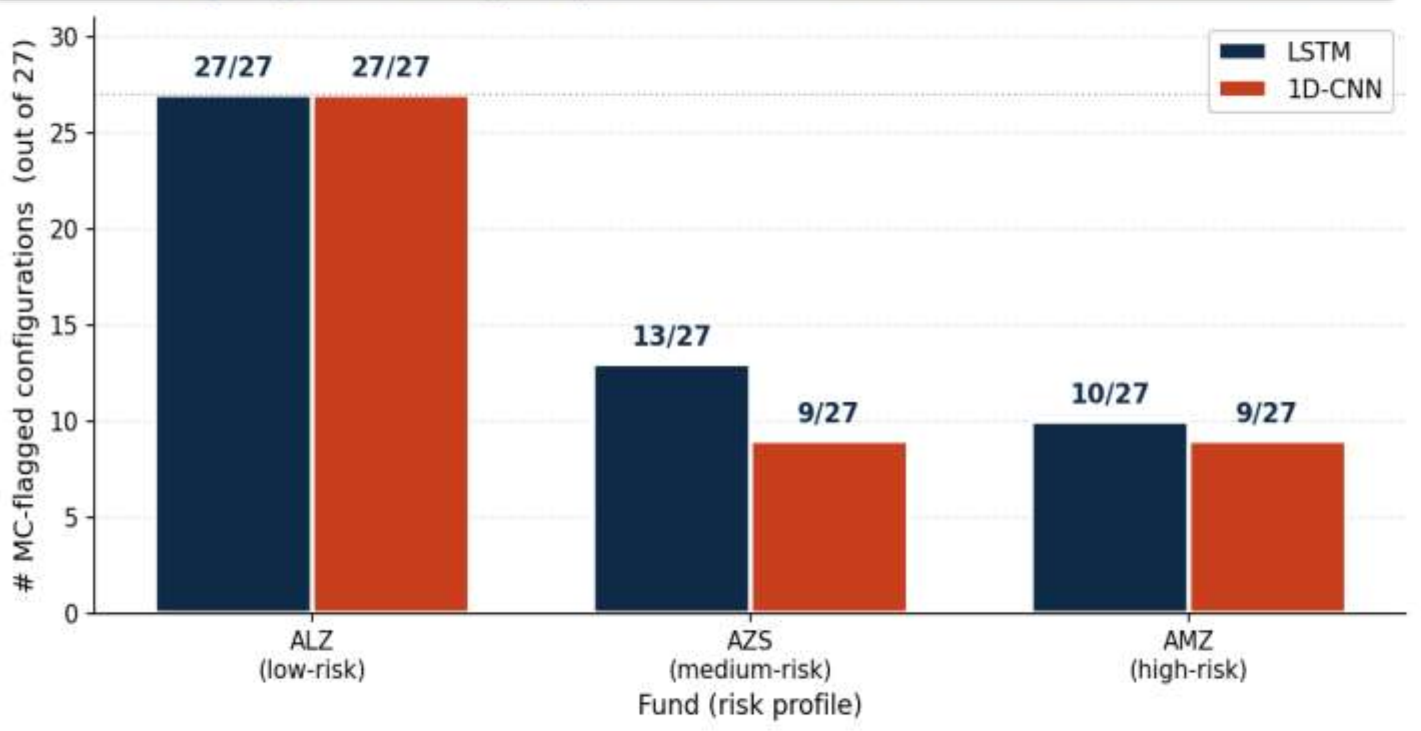
A configuration is flagged as Majority-Class when **Spec = 0** or **Sens = 0** or **Spec = NA** — the model picks one class on every test sample.

MC RATES BY FUND × MODEL

ALZ • LSTM	**27 / 27**
ALZ • CNN	**27 / 27**
AZS • LSTM	**13 / 27**
AZS • CNN	**9 / 27**
AMZ • LSTM	**10 / 27**
AMZ • CNN	**9 / 27**

Finding 1 — 1D-CNN Resists the MC Trap

Figure4 : MC Collapses per Fund



KEY CONCLUSION

Across AZS and AMZ, 1D-CNN's convolutional inductive bias delivers fewer collapsed predictions than LSTM — a robustness, not just accuracy, win.

AT A GLANCE

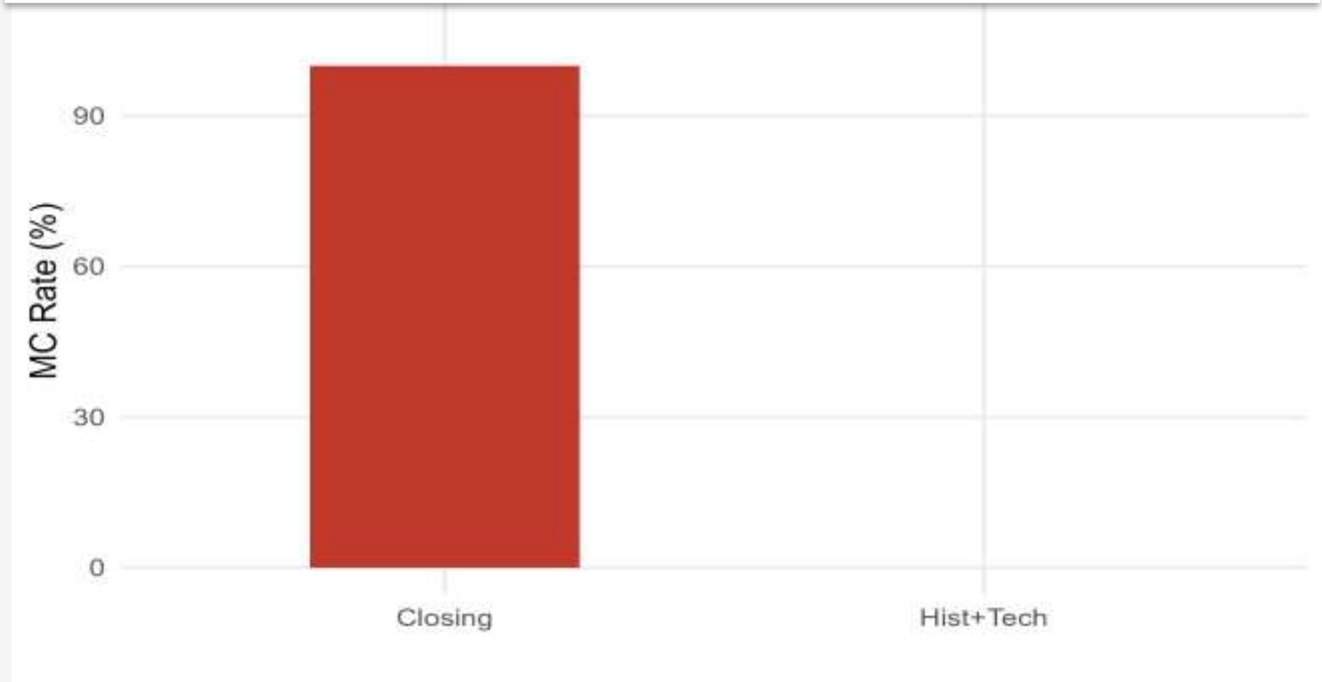
AZS: CNN **9/27** vs LSTM **13/27**

AMZ: CNN **9/27** vs LSTM **10/27**

MC drop: ~**15 pp** on AZS

Finding 2 — Feature Set Drives Performance

Figure5 : MC Rates for Features



OBSERVATIONS

full

best for AMZ LSTM — only configuration where MC = 0/9 with the full set

technical

best for AZS 1D-CNN — RSI + MACD + Momentum lifts BA to 76.70 %

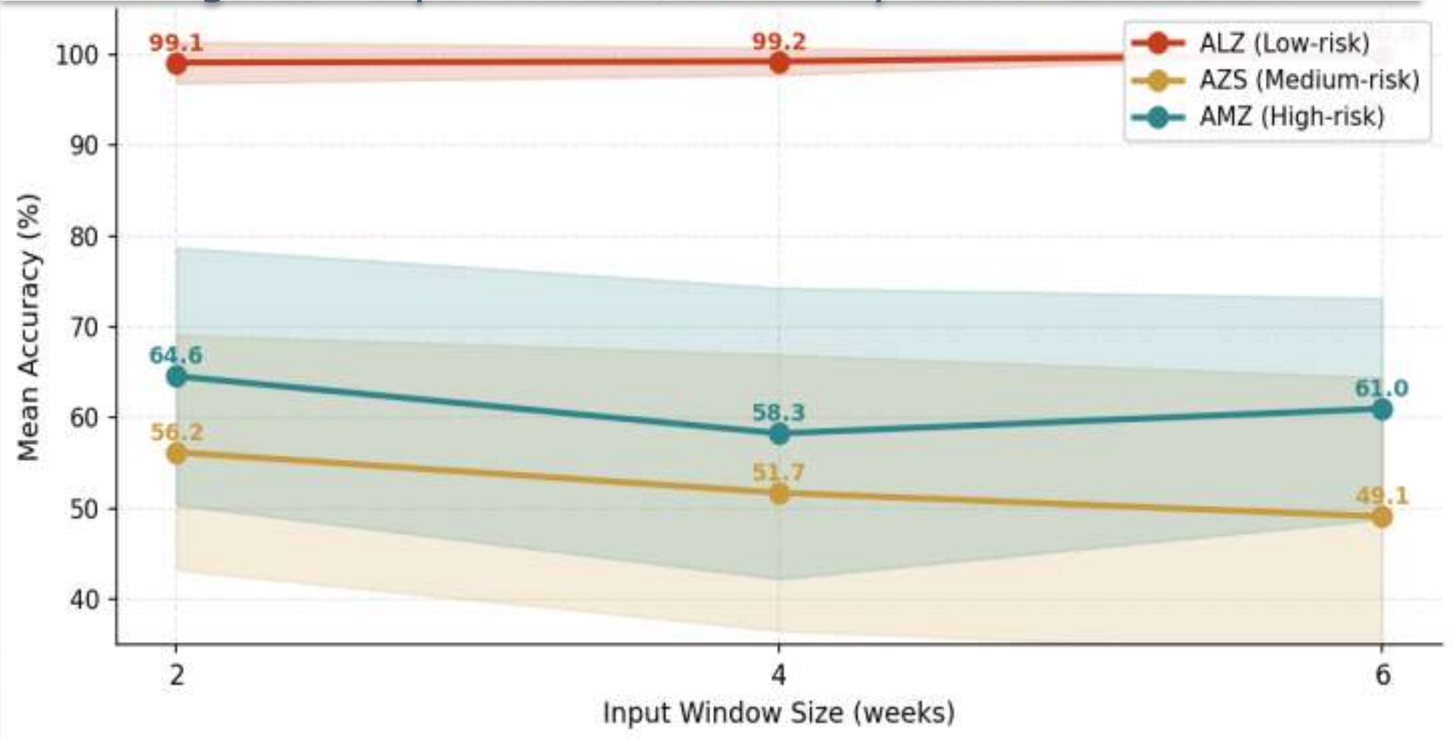
closing

fragile — collapses to MC under low volatility

Feature engineering ≠ optional

Finding 3 — Input/Output Window Sensitivity

Figure6 : Input Window Sensitivity LSTM+CNN Pool



OPTIMAL WINDOWS

AMZ • LSTM

(2, 3)

Mean Acc 80.21 %

AZS • CNN

(4, 3)

Mean Acc 75.56 %

ALZ

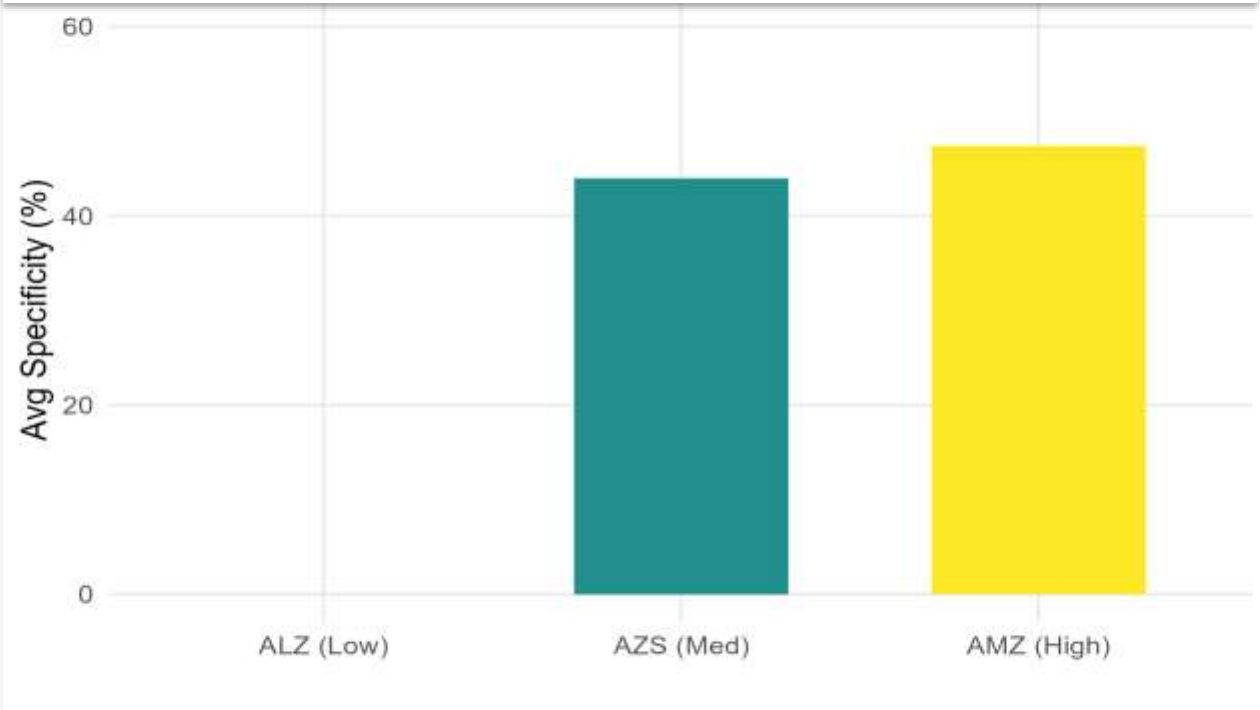
—

Any (collapses to MC)

h = 3 weeks dominates

Finding 4 — Risk Profile Predicts Learnability

Figure7 : Specificity



INTERPRETATION

Low-risk funds (ALZ) are nearly monotone — their direction is degenerate, so deep learners memorise the trend rather than learn it.

Higher volatility (AMZ) creates **genuine class balance**, giving the model something to discriminate.

Bottom line: *directional ML on BES is most useful where conventional advisers say it is hardest.*

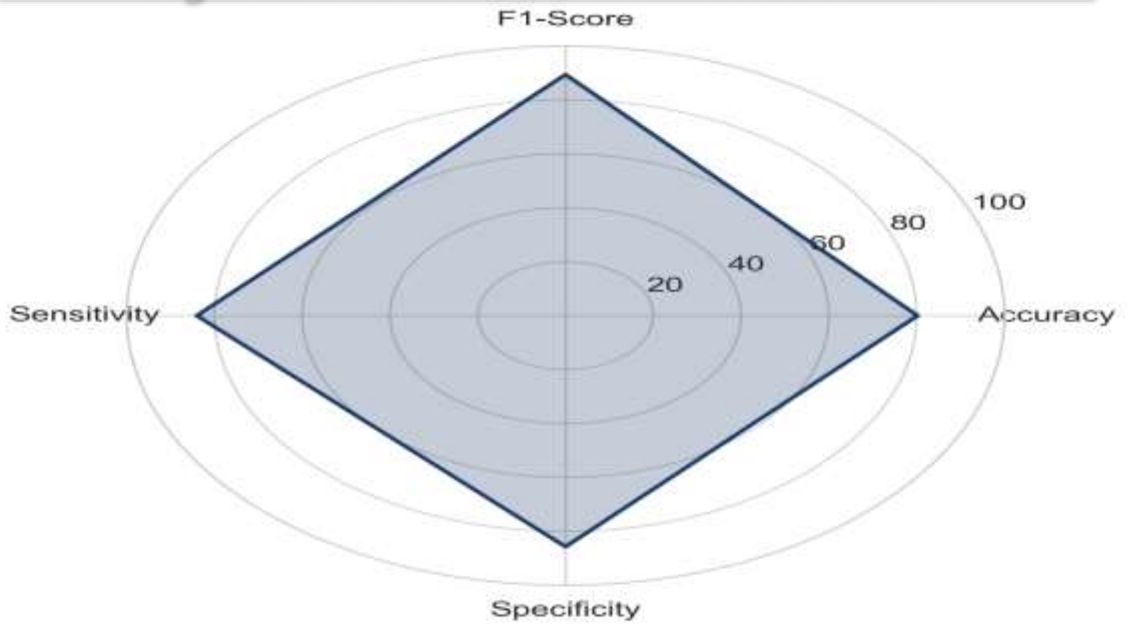
Champion Configurations

Two statistically-significant directional learners

Champion 1 — AMZ • LSTM • full • (2, 3)

80.21 %	84.86 %	89.36 %	0.0001
Mean Accuracy	Balanced Accuracy	F1-Score	p-value (binomial)

Figure8 : AMZ LSTM Excellence Radar



PER-METRIC BREAKDOWN

Sensitivity 84.00 % • **Specificity 85.71 %**

BA = (Sens + Spec) / 2 = (0.84 + 0.8571) / 2 = 0.84855 → 84.86 %

Seed accuracies: 0.8438 / 0.7188 / 0.8438

SD = 7.22 % across seeds *(transparently reported)*

HYPERPARAMETERS

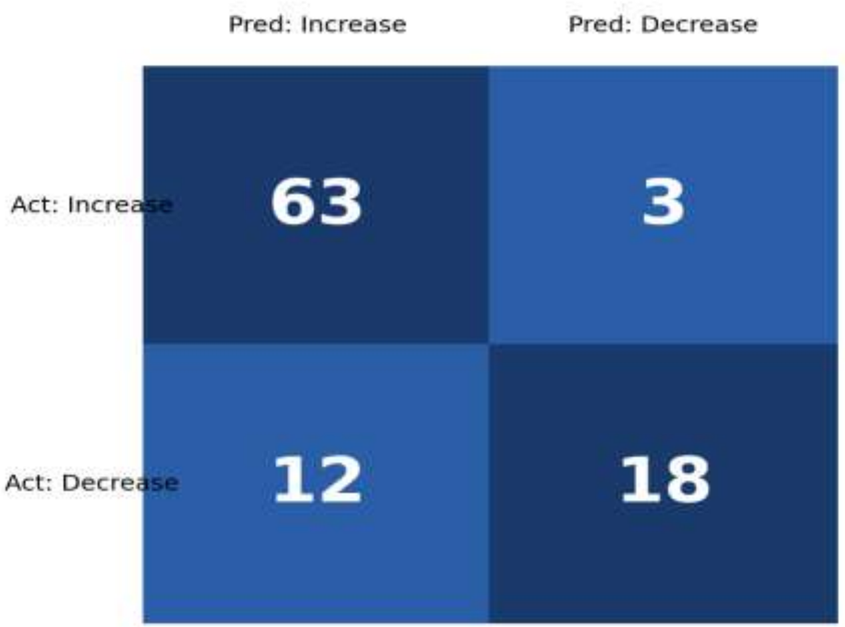
Optimizer Adam • **Activation** Tanh • **Dropout** 0.4

Epochs 50 • **Batch** 16 • **Patience** 5

Feature set: full (7 vars) • class_weight balanced

Champion 1 — Confusion Matrix & Reporting

Figure9 : AMZ LSTM Confusion Matrix



POOLED MATRIX (N = 96)

TP = 63 TN = 18

FP = 3 FN = 12

Diagonal: 63 + 18 = 81 → Pooled Acc = 81/96 = **84.38 %**

REPORTING NOTE

Two valid summaries of the same evaluation:

Pooled accuracy = 81 / 96 = 84.38 % (concatenated test sets)

3-seed mean accuracy = 77 / 96 = 80.21 % (mean of {0.8438, 0.7188, 0.8438})

We adopt the **3-seed mean** as the headline Figure :.

Pooled is exposed for full transparency — never as the headline.

Champion 2 — AZS • 1D-CNN • technical • (4, 3)

75.56 %
Mean Accuracy

76.70 %
Balanced Accuracy

88.89 %
F1-Score

0.0002
p-value (binomial)

INTERPRETATION

Sens 90.91 % • Spec 62.50 %

BA = (0.9091 + 0.625) / 2 = 0.76705 → 76.70 %

Asymmetric strength: **strong recall on "up"** weeks but weaker discrimination on "down" weeks — typical for medium-volatility mixed funds.

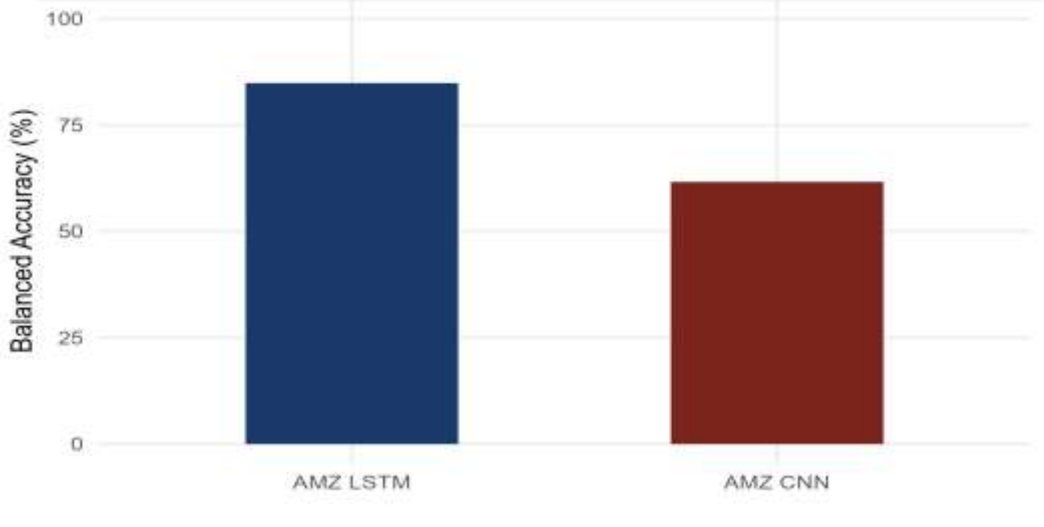
Hyperparameters

Filters 64-128-256 • Kernel 3 • Dense 64 • Dropout 0.2

Feature set **technical** (RSI, MACD, Momentum) • Window (4, 3)

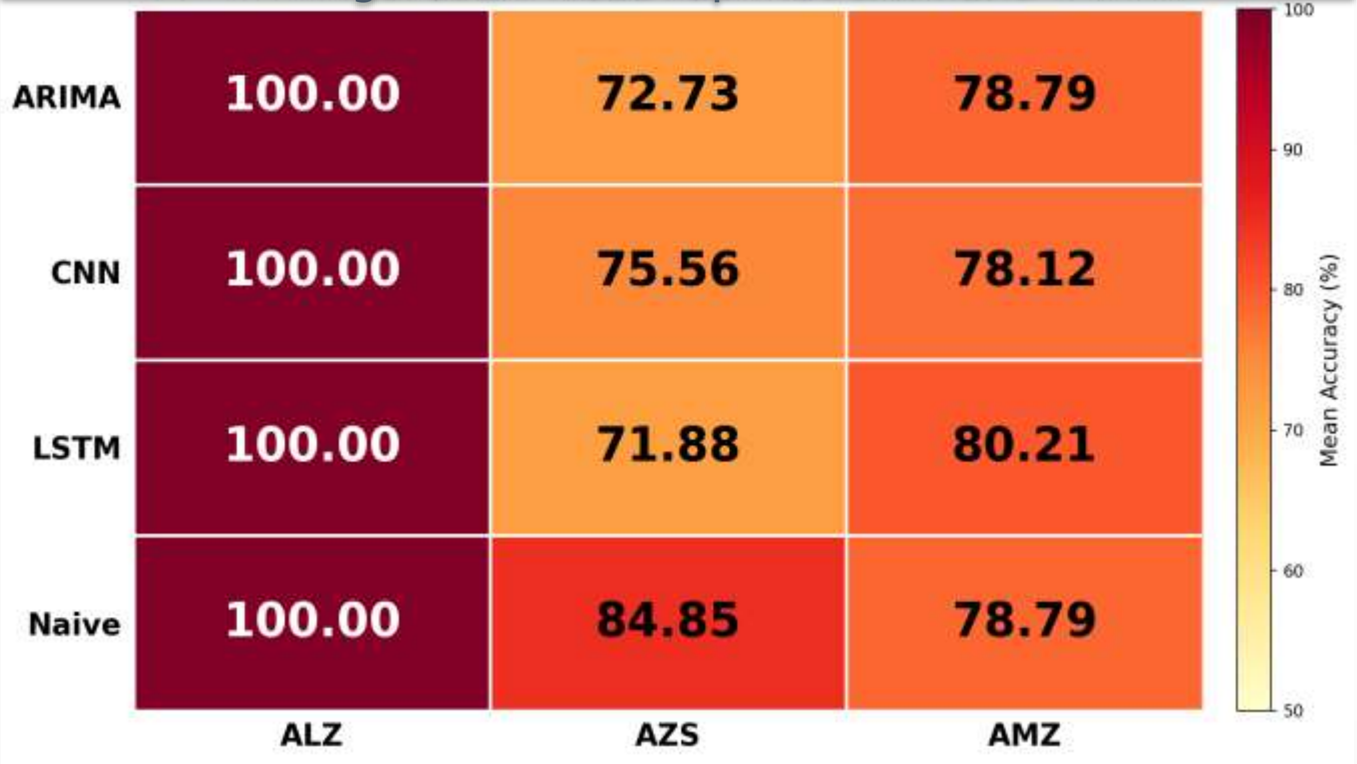
Seed accuracies 0.7000 / 0.7333 / 0.8333 • SD 6.94 %

Figure10 : Balanced Accuracy Comprasion



Cross-View — Model × Asset Performance Heatmap

Figure11 : Heatmap Model x Fund



READING THE HEATMAP

Each cell = best Mean Accuracy of any configuration of that (Model, Fund) pair.

ALZ column = 100 %: pathological monotone — every model collapses to majority class.

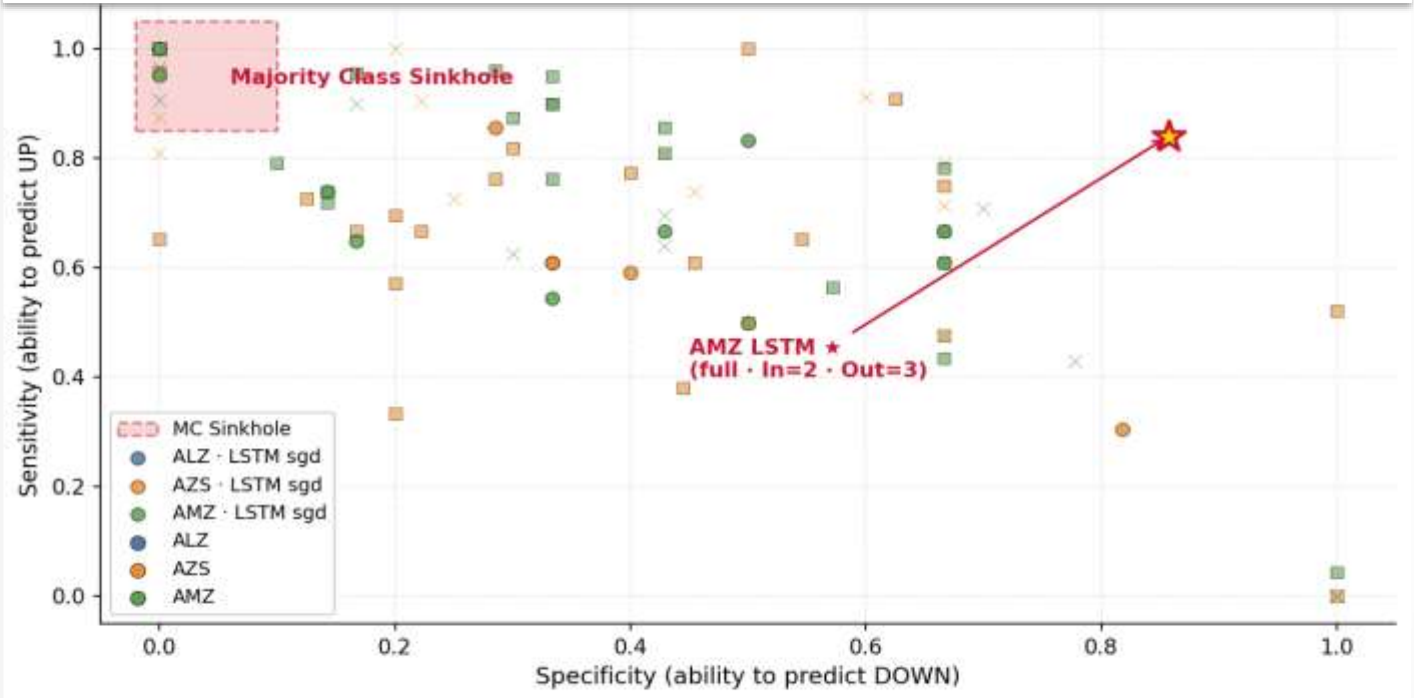
AMZ × LSTM = 80.21 %: only configuration above its Out=3 naive (78.79 %).

AZS × Naive = 84.85 %: persistence dominates — CNN champion 75.56 % falls below.

ARIMA: dominated by naive on AZS, tied with naive on AMZ.

Decision Quality Map — Escape from the MC Sinkhole

Figure12 : MC Sinkhole

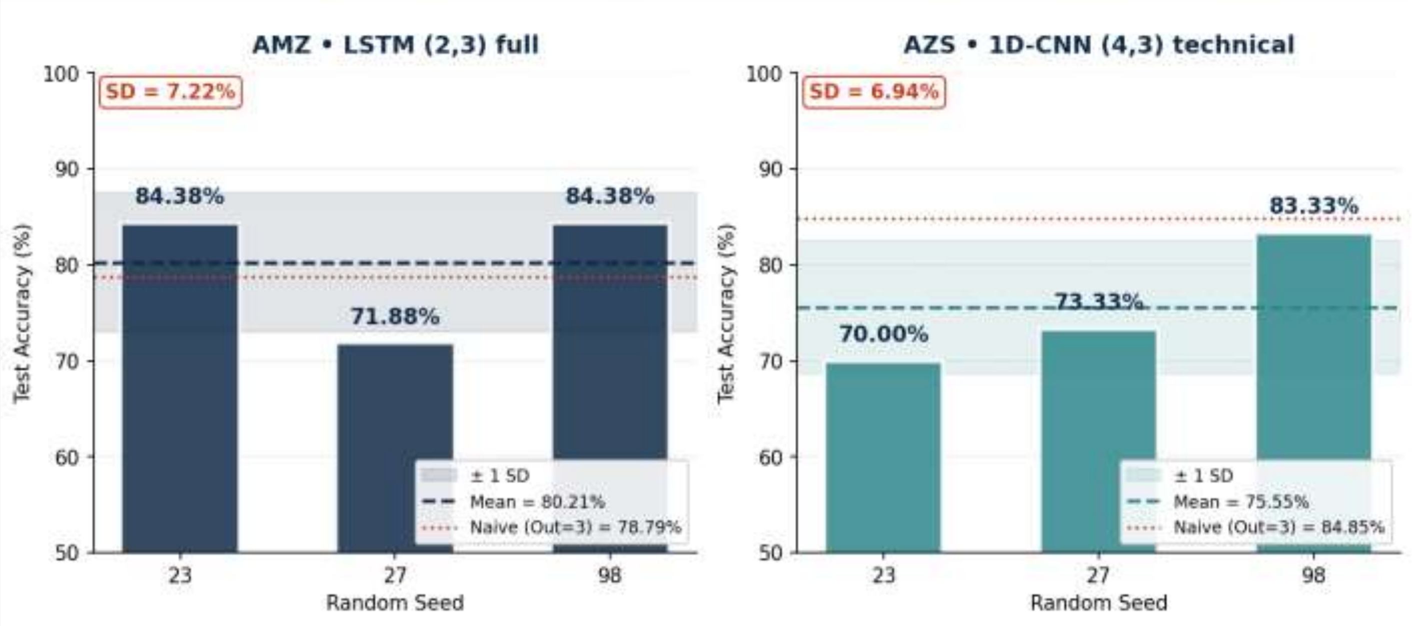


READING THE MAP

- X-axis: Specificity (ability to predict DOWN).
- Y-axis: Sensitivity (ability to predict UP).
- Top-left red zone = MC Sinkhole — Sens≈1, Spec≈0 (predicts only UP).
- ALZ collapses into the sinkhole on every model — pathological monotone.
- Healthy learning lives in the upper-right quadrant (Sens>0.5 AND Spec>0.5).
- AMZ LSTM (full · In=2 · Out=3) ★ escapes to the champion zone — true bidirectional learning.
- Take-away: high accuracy alone is not learning; only points outside the sinkhole are honest predictions.

Seed Stability of Champions

Figure13 : Champion Configurations Across 3 Seed



STABILITY NOTES

AMZ LSTM: 84.38 / 71.88 / 84.38 % → SD = 7.22 %.

AZS CNN: 70.00 / 73.33 / 83.33 % → SD = 6.94 %.

Both means clear the random-guess line (50 %) at $p < 0.001$.

note: *only AMZ LSTM exceeds its Out=3 persistence-naive (78.79 %).*

3-seed 23 · 27 · 98

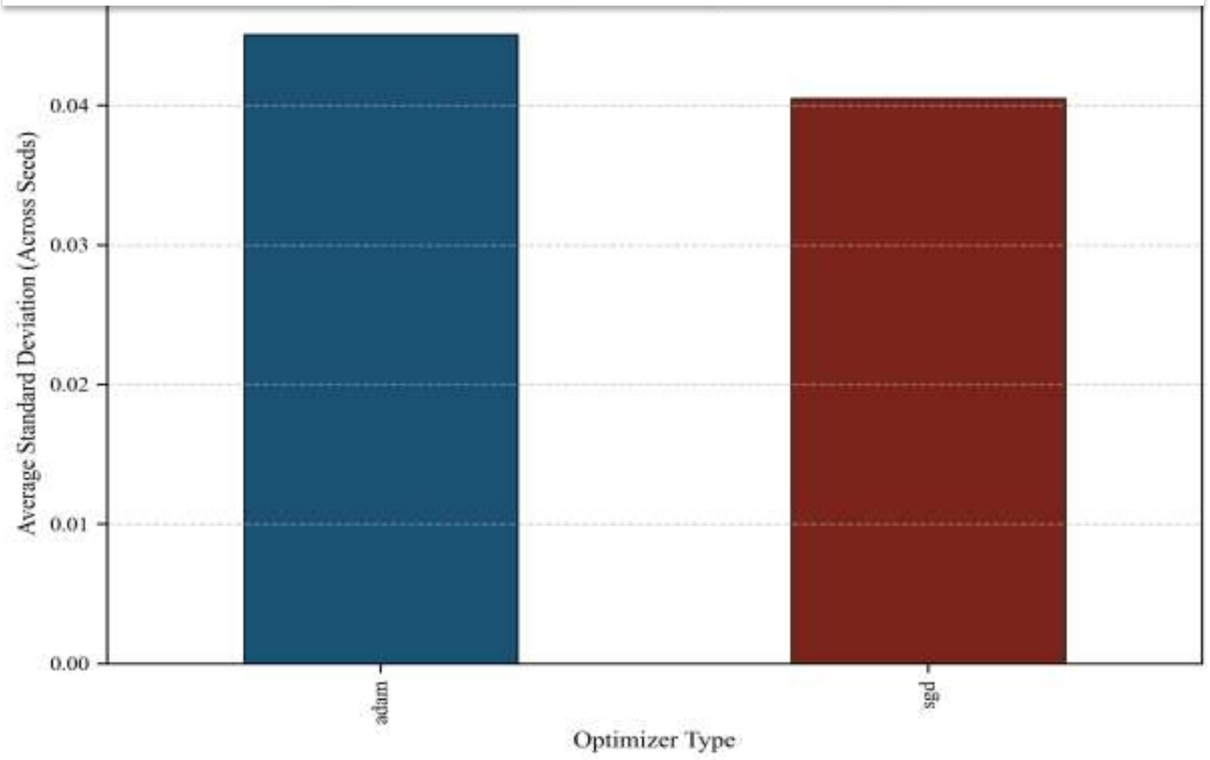
weighted class_weight balancing

$p < 0.001$ champion significance

CSS-fit ARIMA stable

Optimizer & Activation Sensitivity

Figure14 : Optimizer Stability



CONCLUSION

Adam + Tanh wins for the AMZ champion — gradient stability matters under high seed-variance.

SGD competitive on AZS but no statistically significant lift over Adam.

Dropout 0.4 — high regularisation helps the small-sample BES regime.

ReLU collapsed several configurations to MC — inactive units on near-monotone targets.

Limitations & Reporting

Sample size

~260 weekly observations across 5 years — small-sample variance dominates.

Pooled vs Mean

Pooled accuracy (84.38 %) > 3-seed mean (80.21 %). Mean is the headline; pooled is fully disclosed.

Single market

TEFAS only; cross-market transfer to NASDAQ/EuroStoxx left for future work.

Cost-aware loss

Class_weight balancing applied, but not full asymmetric utility (e.g., max-drawdown-aware loss).

Conclusions

01

1D-CNN is more MC-resistant than LSTM

On AZS and AMZ, convolutional bias yields ~15 pp fewer collapsed configurations.

02

Risk profile = learnability ceiling

ALZ (low-risk) is monotone → 27/27 MC across both deep models. AMZ is the clearest signal.

03

Two statistically significant champions

AMZ LSTM (2,3) full → BA 84.86 %, $p<0.0001$ • AZS CNN (4,3) tech → BA 76.70 %, $p=0.0002$.

04

Reporting is non-negotiable

Mean accuracy as headline, pooled CM disclosed, MC defined broadly (Sens=0 OR Spec=0 OR Spec=NA).

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Thank you!

QUESTIONS & DISCUSSION

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